

REVIEW

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Artificial intelligence-based personalised learning in education: a systematic literature review

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Abstract

Personalised learning models can assist students in meeting their unique requirements and objectives for expanding their knowledge, perspective, abilities, and understanding of the educational system. With the advancement of artificial intelligence, technological integration has the potential to play a critical role in personalising the learning experience. This work attempts a systematic review that examines the influence of personalised learning enabled by artificial intelligence on the educational system. The review synthesises the latest research literature (January 2015–June 2025) and includes a total of 125 studies that achieved our inclusion criteria to demonstrate how technology could successfully modify the learning system. Thoroughly examining the role of artificial intelligence-based personalised learning in education is the main focus of this study. In particular, we categorise applications and review algorithms that support personalised learning, compare their approaches, and identify their impact on teaching, learning, and assessment practices across education sectors (K–12, higher education, and institutionally supported online learning). This work contributes to the field of education by providing a much-needed summary of the current state of research and the various opportunities and challenges in this area.

Keywords Personalized learning, Artificial intelligence based personalised education, Artificial intelligence in adaptive learning, Recommender system in education, Generative AI in education

1 Introduction

Personalised learning has been a topic of investigation in the education system for a considerable duration. As artificial intelligence (AI) technology advances and modern personalisation becomes more prevalent, it appears that the applications of personalised learning have changed significantly. Additionally, the potential of generative AI models in educational settings has attracted considerable attention [1]. Despite the availability of various resources and publications on personalised learning, a comprehensive review that focuses explicitly on AI-based personalised learning and its impact on the education system can contribute to further advancements in this field. To address this demand, this



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article analyses the most recent applications of AI-based personalised learning systems, the newest technologies and approaches for these systems, and their impact on K-12 and higher education. We will commence by providing definitions for the terms “personalisation,” “learning,” and “personalised learning” as outlined below.

Personalisation is the act of tailoring or adapting, which refers to the customization of various items such as products, services, or content, to name just a few [2]. In contemporary vernacular, personalisation entails the adaptation of an electronic medium, consumer product, consumer service, or consumer content to suit the user’s personal information or intent [2]. Following Bartolomé et al [3], the aim of personalisation in the education system has been to personalise the learning and teaching process by creating a flexible framework that caters to each student’s individual needs. Learning is defined as the development of a person’s knowledge and abilities, which is a fundamental human activity influenced by personal experiences, ability to process information, attitudes, and environment [4]. Learning is facilitated through interactions with others and personal experiences [4]. Personalised learning is based on the fundamental assumption that each student is unique and has a distinct way of learning [5]. In education, personalised learning is an approach that endeavours to tailor the learning experience to the specific requirements, objectives, and abilities of each individual, which can be facilitated through the use of contemporary instructional technology that offers distinctive educational experiences in a variety of learning contexts [6]. Personalised learning should involve the acquisition of knowledge, abilities, behaviours, and strategies that are meaningful and pertinent to the learner and motivate them to pursue further learning [2]. According to Tadlaoui et al. [2], personalised learning can be characterised by three core elements: the way in which learning content and resources are presented to students, the sequencing and pacing of learning activities to match learners’ progress, and the guidance or support provided by the system to help learners achieve their goals.

Higher education institutions use instructional technology extensively in E-learning and virtual classrooms, with technologies such as the internet and web applications, cloud computing, interactive applications, multimedia projectors, video conferences, smartphones, and mobile technologies benefiting educators and students [7]. The growing use of technology and data in education has been associated with the development of disciplines such as learning analytics, educational AI, and pedagogical data mining, with a particular emphasis on how data may be utilised to research and inform education [8]. One of the primary aims of this research is to investigate the role and purpose of AI-powered personalised learning in the education system. AI is a field of study encompassing the inventions and advancements resulting in computers, robots, and other AI possessing human-like intelligence, as defined by cognitive capacities, training, adaptability, and decision-making skills [9]. Interest in the use of AI algorithms and systems in education is growing [9]. AI has the potential to significantly impact E-learning systems, with some AI-based educational platforms providing an innovative approach to educational administration by reducing the burden on teachers through the provision of feedback platforms for learners [10]. Additionally, various simulation platforms can assist instructors in creating more effective instruction materials by employing simulations instead of text-based lecture notes [10]. The potential advantages of personalised Augmented Reality (AR) instructions in learning also result from the personalised aspect of the instructions and the offering of the instructions through AR technology [11].

We explicitly introduce K–12, higher education, and e-learning contexts because AI-based personalised learning may have distinct roles, adoption challenges, and pedagogical implications across these educational levels. Including them provides the necessary context for comparing applications and impacts throughout this review. The presence of specialised research that can adequately analyse the effects of AI-based personalised learning on the education system can significantly aid in the progress of research in the education system. This systematic literature review article explores the novel algorithms and applications of AI-powered personalised learning and analyses their role and influence within the educational system on the K-12 and higher education sectors. After describing our research method, we first discuss the role of AI-based personalised learning in education. Following this, two major aspects are discussed: the cutting-edge approaches for personalised learning and the novel applications. Furthermore, we conclude this paper by examining the influence of this technology on the education sector, along with discussing the opportunities and challenges that exist in this field.

As indicated in Fig. 1, this systematic literature review's pipeline and all subsections are intended to address four research questions.

- **Research Question 1:** What conceptual and theoretical role does AI-based personalised learning play in education (what it does, why it matters, and how it is defined)?
- **Research Question 2:** What techniques have been proposed to implement AI-based personalised learning?
- **Research Question 3:** What applications of AI-based personalised learning can be identified to address different educational needs?
- **Research Question 4:** What impact has AI-based personalised learning had on teaching, learning, and assessment practices?

The following are the ways in which this review study contributes to the AI-based personalised learning literature:

1. **Expansive Analysis:** This review analyses an extensive range of recent research to offer a comprehensive overview of the progress made in educational technology for personalized learning over the past decade.

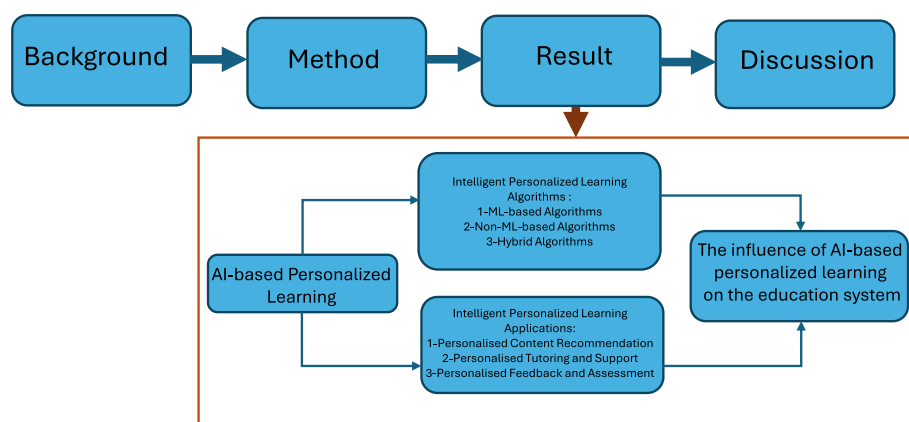


Fig. 1 A schematic overview of the pipeline of this systematic literature review for AI-based personalised learning in education system

2. **Comparative Analysis of AI Applications and Algorithms:** This study assesses and contrasts various AI-driven tools and methodologies, emphasizing their unique applications and efficacy in personalised learning environments.
3. **Detailed Examination of Educational Impact:** This work evaluates the various influences of AI-based personalized learning across primary, secondary, and higher education, describing how AI shapes educational outcomes and dynamics within these settings.
4. **Opportunity and Challenge Identification:** The study outlines future research possibilities and technological developments in personalized learning while also discussing the challenges that must be addressed.

2 Method

Our systematic review follows the principles in [12, 13] and has been conducted to extract meaningful information from the existing research based on AI-enabled personalised learning systems in education. Our review involves establishing research questions, identifying the corpus of search terms, choosing databases, generating justifiable inclusion and exclusion criteria, evaluating existing novel approaches and applications, expanding search results, realising meaningful outcomes from AI-based personalised learning's influence on the education system, and discussing contributions.

2.1 Literature search and screening procedures

We determined relevant articles for our meta-analysis utilising bibliographic databases and publications in personalised learning. The initial search was conducted across multiple search engines and databases, including Web of Science, ERIC, Google Scholar, Scopus, Sage Journals, and Springer Journals. The following Boolean search string was applied: ("personalised learning" OR "personalized learning" OR "customised learning" OR "customized learning" OR "adaptive learning" OR "tailored learning" OR "individualised learning") AND ("AI" OR "artificial intelligence" OR "machine learning" OR "deep learning" OR "generative AI" OR "recommendation system" OR "technology-based learning" OR "e-learning") AND ("education"). During the screening stage, we restricted the selection to English-language, peer-reviewed journal articles and conference papers published between January 2015 and June 2025. Details of the inclusion and exclusion parameters are provided in Section 2.2 and summarised in Table 2. The initial main search generated 371 publications. The 44 duplicate results were eliminated from the analysis. The titles and abstracts of the remaining 327 papers were evaluated based on the inclusion criteria of this study. During the screening process, 86 articles were omitted. The entire texts of 241 papers were reviewed, and 125 were considered relevant to the goal of this review. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework guided the selection and analysis of literature for this systematic review. Figure 2 presents the PRISMA flowchart that summarises the study identification procedure. Table 1 summarises the search parameters for this systematic review.

2.2 Inclusion and exclusion criteria

Inclusion and exclusion criteria have been utilised to restrict the pool of selected articles that resulted from our search. Articles had to be published in English between 2015

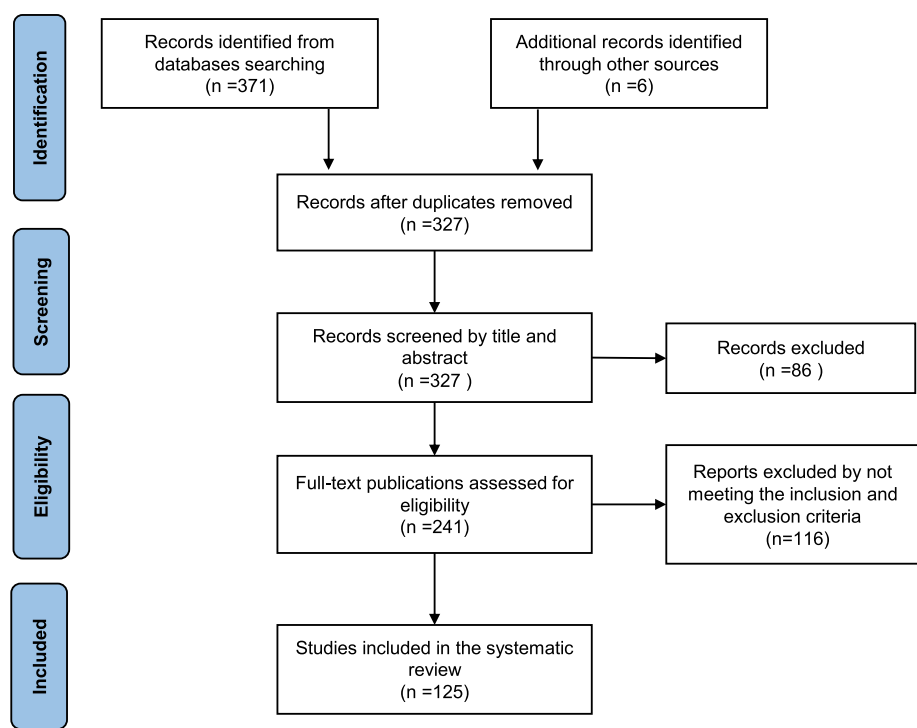


Fig. 2 PRISMA diagram illustrating study selection methods (n = number of articles selected in this review)

Table 1 The search parameters for this systematic literature review

Search parameter	Details
Research questions	Established four main research questions
Corpus of search terms	Personalized learning, education system, artificial intelligence, customized learning, adaptive learning, recommendation system, individualized learning, machine learning, tailored learning, e-learning, AI-based learning, technology-based learning, generative AI, deep learning
Databases	Web of Science, ERIC, Google Scholar, Springer Journals, Scopus
Search dates	January 2015–June 2025
Initial search results	371 publications
Total relevant publications	125 publications

and 2025 from prestigious journals and conferences to qualify for inclusion, and studies had to investigate AI-based personalised learning in the education system. The abstracts and introductions of selected articles were examined to assess if the authors described the purpose of the research. In addition, the authors’ solutions to the research question were evaluated in the method and results section. To ensure the quality of the included work, we excluded papers whose methods for personalised learning were assessed to be irrelevant to artificial intelligence techniques. In addition, we excluded studies where the application of their technique for the purpose of personalised learning could not be determined. We concentrate on publications published in high-quality scientific journals and conferences on the subject of education and e-learning. This review, therefore, excludes unpublished dissertations, research reports, non-English studies, and studies that were not adequately reported. Table 2 summarises the inclusion and exclusion criteria. Table 3 lists the journals and conferences considered in this review and the total number of papers from each of these sources.

Table 2 Inclusion and exclusion criteria for this systematic literature review

Inclusion criteria	Exclusion criteria
Articles published in English	Non-English studies
Published between January 2015 and June 2025	Unpublished dissertations and research reports
Studies investigating AI-based personalised learning in the education system	Studies not adequately reported
Published mostly in prestigious journals and conferences	Methods for personalised learning that were assessed to be irrelevant to AI techniques
The clearly defined research questions in the abstract and introduction	Studies where the application of their technique for the purpose of personalised learning could not be determined
Described AI-based personalised learning solutions in the method and results section	

Table 3 Result-set size for the number of papers on “AI-based Personalised Learning” from January 2015 to June 2025 by the top journal/conference that has been selected for this work

Journal	Count of papers	Ranking (SJR)	H Index
Review of Educational Research	5	5.772-Q1	186
Computers & Education	7	3.651-Q1	232
Computers and Education: Artificial Intelligence	11	3.227-Q1	29
International Journal of Educational Technology in Higher Education	5	2.578-Q1	61
The Internet and Higher Education	4	2.426-Q1	117
Journal of Computer Assisted Learning	12	1.842-Q1	114
International Journal of Artificial Intelligence in Education	20	1.842-Q1	62
Education and Information Technologies	8	1.301-Q1	76
IEEE Transactions on Education	5	0.79-Q1	76
Others	26		
<i>Conference</i>			
International Conference on Artificial Intelligence in Education (AIED)	9	ERA* Ranking: A	
International Conference on Learning Analytics and Knowledge	4	ERA Ranking: A	
Others	9		
Total	125		

*<https://www.arc.gov.au/excellence-research-australia>

2.3 Data analysis and quality assessment

We employed thematic synthesis to examine the articles in which themes arose from the main studies, allowing us to aggregate trends across various approaches to solving our research questions. For the thematic synthesis, an inductive coding approach was applied. Themes were identified through repeated reading of the studies, with relevant text segments manually coded. The coding was primarily carried out by the first author, while a subset of studies was reviewed by co-authors to check consistency and reduce potential bias. Thematic analysis followed a multi-stage coding process. First, initial open codes were generated inductively from the dataset through close reading of the texts. These codes were then grouped into higher-level categories through iterative comparison, refinement, and merging of similar codes. Finally, overarching themes were derived from the categories to capture the broader patterns in the data. The process of coding and theme development was conducted systematically. Then, the results, methodologies, and influences were individually coded for each publication. To improve

reliability, the coding was checked and agreed upon. New codes were recognised, adjective themes were formulated, and interpretive themes were produced.

We established several criteria for evaluating the quality of included studies, such as how well the article's aims were made explicit, whether or not the proposed methods were sufficiently specified and if the accuracy of the proposed methods was measured and validated. For our systematic review, we found that this evaluation methodology and assessment criteria well accounted for various approaches while still giving a comprehensive assessment of bias risk in the included research.

With regard to quality assessment, we followed established criteria commonly used in education-focused systematic reviews, including clarity of study aims, adequacy of methodological reporting, and evidence of validation or evaluation of results. This approach provided a structured and transparent assessment suited to the interdisciplinary nature of AI-based personalised learning.

3 Result

According to the four defined research questions, the results of the systematic review are presented in four sections: (1) the role of AI-based personalised learning in education, (2) innovative methodologies for personalised learning, (3) novel applications of personalised learning, and (4) the impact of this technology on the education sector.

3.1 AI-based personalised learning

With the evolution of AI technology, it has found applications in a variety of fields, including education. As a result of AI's beneficial effect on education, interest in the application of AI technologies in education has surged. This effect extends to individualised learning as well. Numerous studies conducted in recent years have concentrated on the application of AI technologies in individualised learning systems. In this section, we describe the function of customised learning in education and the role of AI in personalised learning systems. Given the advantages of AI approaches, there is potential for recommending the most suitable educational materials from an extensive array of available resources, providing guidance on a well-planned curriculum, and undertaking accurate evaluations of student performance, among other things [14]. We also discuss the benefits of employing AI in education and highlight how personalised learning systems can be leveraged to personalise learning for individual students, facilitating more efficient learning.

Following Tadlaoui et al. [2], elaborated that individuals receive training to expand their knowledge and abilities in a particular subject while also acquiring new expertise, skills, and insights. Swaggerty et al. [15] mentioned that educators should be assisted in using acquired information with different students in the classroom during their teacher education. Taxonomies like Bloom's have long been used in education to identify the most important aspects of learning [16, 17], and several studies have been conducted in the fields of special education and collaborative teacher education [18]. As discussed by Holmes et al. [19], AI is perhaps the most transformative technology of the first half of the 20th century; it reshaped practically every sector, if not all, of humanity's undertakings. They also noted that governments and corporations worldwide are investing heavily in various technologies. Additionally, they mentioned that AI promises to have a substantial impact when it comes to education. AI-based personalised learning systems,

whether welcomed or not by students and teachers, are increasingly being implemented in educational institutions worldwide and have substantial effects [19]. For instance, AI can simplify a teacher's life and make their job more fulfilling by relieving them of tedious tasks, such as marking tests and affording them more time to interact with their learners [19]. AI has now become an important, widespread, and unavoidable component of our daily lives, although its deployment in education may still be a mystery for many [19]. Moreover, AI is being implemented as a curriculum in certain conventional schools and being developed to enhance online tutoring and is being studied as a means of improving teacher training [19]. Following Maghsudi et al. [14], the development of AI and machine learning and advancements in big data analysis have brought about fresh perspectives that improve customised education in various ways. Maghsudi et al. [14] stated that AI and machine learning techniques enable the educational platform to collect the student's characteristics accurately by examining past experiences and scrutinising the available big data to identify the student's features precisely. This enables the platform to suggest suitable content from several options, recommend a well-planned long-term curriculum, and connect appropriate learners through suggestions, accurate performance assessment, and other means [14].

According to Chen et al. [9], a growing number of educational applications of artificial intelligence transcend the conventional concept of AI as a supercomputer by using embedded computer systems. They also believed that AI is advancing at a breakneck pace, and new applications in education are emerging. Individualised education is the goal of AI-based personalised learning, which employs approaches associated with data visualisation techniques, machine learning, semantics, and learning sciences [9]. For example, AI-based competency learning, which generates crucial student data, can properly identify student perspectives and forecast the relevant competencies they can follow, allowing institutions to respond proactively [9]. Numerous papers explored and illustrated the educational applicability of AI-based personalised learning. In schools, AI can be utilised to assist with management and training. Further, AI can enhance the learning experience of students, encompassing different educational levels ranging from early childhood education to higher education institutions [9]. Such modern systems, which are based on AI and cognitive theory, direct students through each stage of a problem solution by providing recommendations and feedback as required from expert-knowledge repositories [20].

Artificial intelligence in education, according to Xu et al. [21], has the potential to liberate instructors from monotonous mechanical tasks, giving them more time and energy to focus on cultivating students' high-level thinking and creativity abilities, transforming "teachers" into true "leaders" and "coaches". Further, they argued that teachers are no longer just individuals who disseminate knowledge; rather, they have evolved into individuals who assist students in meeting their requirements and growing via personalised learning programs. They also argued that the AI system could give each student a unique learning style, allowing them to learn effectively. AI-powered software like the "Home-work Helper" enables students to review class materials independently [21]. The software allows students to upload images of ambiguous questions, which can be resolved using image recognition technology in artificial intelligence [21]. This goal would be difficult to achieve with a non-AI-based system. Additionally, they stated that today, AI is most frequently utilised in education for image and speech recognition and human-computer

interaction. Tutoring, virtual learning, and instructional practices are just a few of the applications for which these applications can be used. A good illustration of how AI has been applied in the classroom is the intelligent tutoring system [21].

According to Troussas et al. [22], intelligent and adaptable strategies can provide students with a fresh learning experience. They also discussed that intelligent applications could help encourage personalised and collaborative learning in educational systems. Then, following Chounta et al. [23], technology-based customised learning is a component of an intelligent learning system that can significantly enhance education. Further, they believed that recently, AI had been utilised as a tool for students to learn through experimenting with AI algorithms or as a technical tool to enable the personalisation of educational experiences and the adaptation of training to individual learners' needs and objectives. Furthermore, there has been considerable interest in the rise of online teaching tools. The utilisation of AI techniques and the analysis of available big data enables the exploration of learners' characteristics, resulting in the development of a recommendation system that can provide students with the most pertinent material in higher education while also assisting teachers and content designers [14]. Following the work described by Maghsudi et al. [14], we developed Fig. 3 to illustrate an educational system's recommendation system for personalised learning via the application of AI. Student data can be gathered on academic performance through traditional forms of assignments, such as problem sets, computer programmes, and laboratories, as well as through online activities and assessments, which are more commonly used in higher education [14]. To establish the policies following Xu et al. [24], the process can be divided into two stages. In the first stage, a forward-search backwards-induction algorithm is employed to choose candidate course sequence recommendation policies that will result in shorter time-to-graduation. In the second stage, an online regret minimisation learning model is developed to select personalised course sequence recommendation policies from the candidate policies. The aim of this algorithm is to maximize the student's GPA performance [24]. To generate AI-based personalised course recommendations, the algorithm learns from the performance of students with diverse

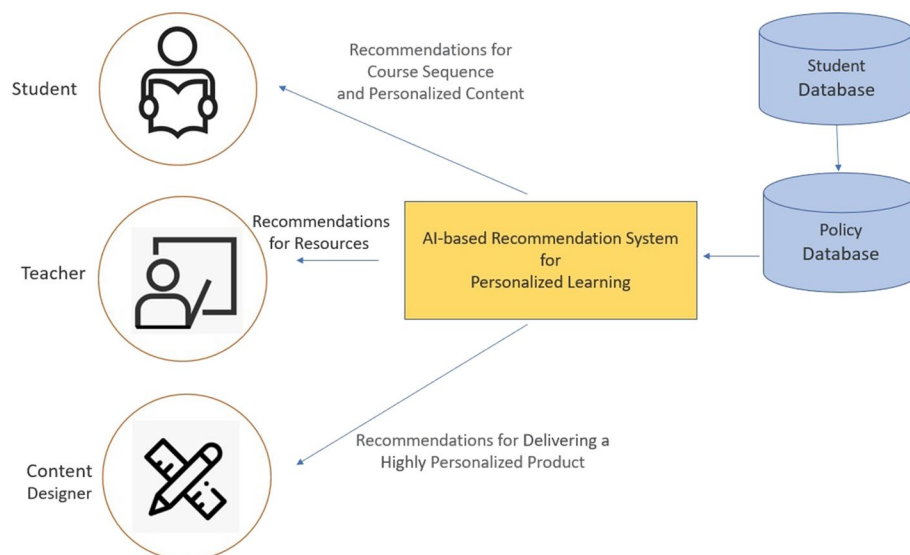


Fig. 3 An AI recommendation system for personalised education system

backgrounds in specific course sequences and suggests an ideal sequence for new students [24].

Having sufficient data in this field might be challenging; for example, in the past, the application of reinforcement learning to develop instructional policies has been hampered by the lack of substantial amounts of actual learner data [14]. Nevertheless, recent techniques have attempted to overcome this limitation by employing synthetic learner models derived from data or cognitive theories to imitate learner data [14]. Additionally, AI generative models offer an alternative solution for the shortage of legitimate data since the generator generates novel material resembling the input data, and the discriminator distinguishes the artificial output from real data [25]. As stated previously, although the AI recommendation system has the potential to provide relevant information to students, teachers, and content creators, this research primarily concentrates on the content recommendation systems that directly provide educational materials to learners, and in some instances, recommendations to teachers. We further expound on this in the Personalised Content Recommendation section below. The advancement of computer systems in terms of processing capabilities and data storage, along with the rapid increase in digital technology usage, led to the development of Big Data [26]. This in turn led to the emergence of the fields of learning analytics and educational data mining, which aim to leverage the vast data available to create new models that promote student learning [26]. These methodologies employ software-based processes such as prediction, clustering, and data extraction for human judgment, as well as social network analysis, to take advantage of the large amounts of data available in e-learning platforms, which were not accessible prior to the advent of AI [26]. The integration of big data and artificial intelligence in education holds great promise as it is expected to provide insights that will enable personalised education for each learner, leading to improved engagement and better learning outcomes compared to the past [26]. Advancements in machine learning have led to the development of AI-generated media, which can produce highly realistic text, images, voice, and video content [25]. This technology can create lifelike representations of characters, including those of historical or fictional figures or deceased relatives [25]. AI-generated characters have promising applications, such as providing personalised educational content and offering health communication and coaching that is tailored to individual needs [25].

Following Shemshack et al. [4], to reveal the number of published articles on AI-based personalised learning, we used Scopus¹ and searched² for publications from 2000 to 2024, using the key terms (“Personalized Learning” or “Personalised Learning”) and (“Artificial Intelligence” or “AI”) and “Education”. Although this study reviews literature from January 2015 to June 2025, we limited the number of published articles to 2024, as the current year (2025) is still in progress (as of June 2025), and the publication data is incomplete. We then created Fig. 4 based on the retrieved data from Scopus, showing that the number of published articles on personalized learning has increased exponentially. This highlights that personalized learning has become a hot topic in AI-guided education. Notably, there was a sharp increase from 2023 to 2024, which might be due to the recent emergence of generative AI technologies such as ChatGPT. Furthermore, as part of our comprehensive literature review, we classified the current personalized

¹ <https://www.scopus.com>.

² Retrieved on 12 June 2025.

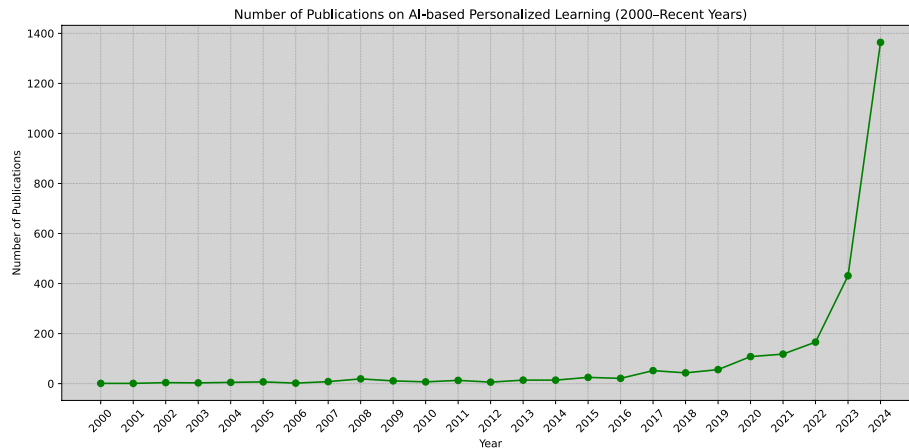


Fig. 4 Number of publications in Scopus with key terms (“Personalized Learning” or “Personalised Learning”) and (“Artificial Intelligence” or “AI”) and “Education”

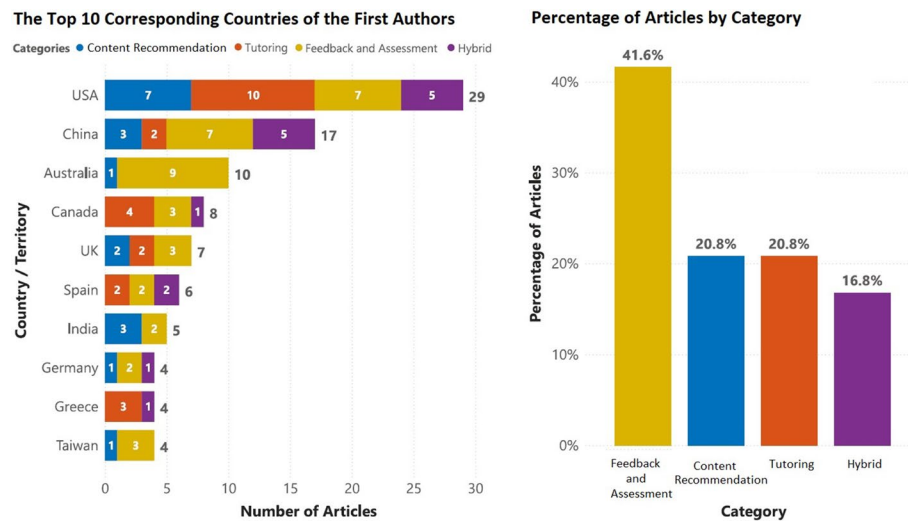


Fig. 5 The specific applications of AI-powered personalised learning systems that the 125 articles used in this literature review focused on, along with the top 10 corresponding countries of the first authors

learning applications into three extensive divisions: content recommendation, tutoring and support, and feedback and assessment. To illustrate the specific applications of the 125 articles examined in this literature review, we developed Fig. 5 to show the distribution of these applications and further list the top ten affiliation countries of the first author. We observe that AI-powered personalized learning research is a hot topic globally. Figure 5 shows that 41.6% of the articles focused on feedback and assessment, 20.8% on tutoring and support, and 20.8% on content recommendation. Additionally, a category labelled “Hybrid” accounts for 16.8% of the considered papers, representing those studies that discussed more than one application classification. Such detailed categorization and geographical distribution provide practical insights into the prevalent research trends and the collaborative nature of this developing field.

3.2 Intelligent personalised learning algorithms

This section investigates novel computer-based methods for tailoring learning to answer the review's second research question. The progress of educational technology has promoted the development of tailored learning algorithms [27]. By analysing implicit user input, technology-based detection approaches are widely utilised to increase the collection of personal learning data in the learning area [28]. Computer-based detection approaches use a variety of digital algorithms or programmes to automatically assess a person's individual characteristics. As a result of these activities, the learner model's knowledge about the learner is often updated [28]. Because it adapts rapidly to changes in the learner's personal characteristics, the technology-based detection strategy is regarded as more reliable than the questionnaire method [28].

In this section, we review recent computer-based methods and strategies developed for tailoring learning experiences. Following Normadhi et al. [28], the techniques are categorised into three types of technology-based detection strategies used to identify personalised learning: machine learning, non-machine learning, and hybrid approaches. Following the works that have been described by [29–31], we produced Fig. 6 to illustrate the link between machine learning and non-machine learning methods to AI, in which deep learning and natural language processing are subsets of machine learning, whereas machine learning is a subset of AI [29].

3.2.1 Machine learning-based algorithms

Machine learning could be defined as a process that makes use of an algorithm to make predictions based on prior training and learnings [32]. Machine learning is a broad term that encompasses both supervised and unsupervised learning techniques [32]. Deep Learning is a relatively new field of research in machine learning that was established with the purpose of bringing machine learning further to one of its primary objectives, AI [32]. Table 4 illustrates some examples of machine learning algorithms that have been used to produce intelligent personalised learning. To develop an interactive platform that integrates gesture recognition technology, Junokas et al. [33] focused on establishing intuitive and tailored learning environments via the use of machine learning techniques.

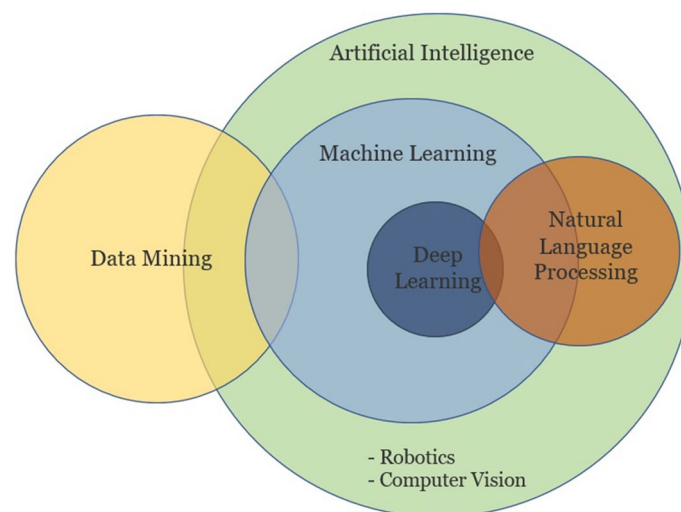


Fig. 6 The relationship between machine learning and non-machine learning techniques with AI

Table 4 A summary of the samples of machine learning-based algorithms for the development of intelligent, personalised learning is addressed in this study

Type of algorithm	References
Nested Cross-Validation	Cross-validation cycles as a standard machine learning method [39]
Random Forest Regression	Using a supervised machine learning methodology (random forest regression) [38]
A Feature-based Machine Learning	Using FACE, a supervised feature-based machine learning technique [40]
Hierarchical Clustering	Utilising hierarchical clustering with Ward's technique [41]
Agglomerative Hierarchical Clustering	Clustering based on Ward's algorithm [42]
Bayesian Networks	Flowchart-based intelligent tutoring system that uses Bayesian networks [43]
Natural Language Processing	Providing personalised hints and feedback using Natural Language Processing [44], Providing sequence of research [45]
Hidden-Markov Model	Using a hierarchical hidden-Markov model for recognition of personalised gestures [33]
Deep Learning Model	Deep learning-based Recommendation of personalised research papers [34], Deep learning technique for detecting hand-over-face movements in photos for improving intelligent tutoring systems [35]
Convolutional Neural Network (CNN)	Levy flight distribution (LFD), which is based on CNN [46]. Employing CNN to identify the learner's personality to help personalized learning [47]
Recurrent Neural Networks	Generating real-time feedback using supervised machine learning techniques by recurrent neural network classifiers [48]
Long Short-Term Memory Networks	Using long short-term memory Neural Networks [49]
Reinforcement Learning Algorithm (RLA)	Using RLA to offer personalised guidance on the most effective Self-Regulated Learning (SRL) strategy [50]
Generative AI like ChatGPT	Utilising ChatGPT as a supportive tool for educational purposes, which can provide personalised and flexible learning experiences [51]. Determining ChatGPT's impact on personalised feedback and interactive learning [52]. Deep generative models [53]

They used the Hierarchical hidden Markov model for gesture detection in order to construct a minimally developed yet adaptive model for tailored user interactions. Following them, while the Hierarchical hidden Markov model is effective in doing typical machine learning tasks, it is particularly adept at modelling movement characteristics.

As Hebatallah et al. [34] explained, deep learning approaches are essentially several non-linear layers of artificial neural networks. They believed that a significant advantage of deep learning is its ability to analyse and learn from enormous volumes of unsupervised data, which makes it an ideal tool for big data analytics since the raw data is mainly unstructured. Following them by utilising deep learning approaches to derive important representations of the data, it is feasible to get a semantic and relational knowledge of the material included within such large-scale textual data. Hebatallah et al. [34] recommended appropriate research papers using deep learning algorithms based on the inherent meaning of the papers that matched the user's interests. Additionally, they stated that user profiles might be created based upon certain basic information provided by the student, such as the articles the learner is interested in and input via a graphical interface. Additionally, they argued that the profiles would be enhanced utilising user interaction data taken from users' logs. Another example of a deep learning approach is the research study of Behera et al. [35]. They investigated the automatic identification of nonverbal behaviours such as hand-over-face motions, head and eye movements, and emotions associated with facial expressions throughout the learning process. They suggested a new deep learning technique for automatically detecting hand-over-face

movements in photos, which might be used to improve intelligent teaching systems. In addition, Ren et al. [36] also used deep learning algorithms to anticipate the user's subsequent interaction and the time of that interaction, enabling online learning platforms to offer more personalised and efficient services.

Casey et al. [37] used educational analytics and a machine-learning algorithm to predict student's performance by examining the rate at which students apply cognitive responses throughout a semester. They provided a basic classification method for identifying weak performers early on and demonstrated how it might be improved by integrating data on when students first utilise a topic. They created a binary classifier that estimates a student's success on the course's final written test (pass/fail) using a variety of factors. In addition, for handling the form and the context of personalized writing, Beigman et al. [38] examined the extent to which a writing example may be automatically rated for expression of utility value using supervised machine learning methods. They built prediction models employing linguistic characteristics, specifically, automatically identified indicators of different language qualities of the writing samples, using random forest regression. Regarding Fratamico et al. [39], collaborative simulations can aid in the process of inquiry-based learning. They argued that, however, like with other exploratory education settings, these unstructured settings may not necessarily promote successful learning. They believed that offering customised support has the potential to significantly aid in the improvement of student learning through these rich activities. Following them, flexible support needs a student model that is capable of evaluating learning and informing appropriate feedback. Fratamico et al. [39] suggested a framework for modelling students in interactive simulations in order to solve the difficulty of modelling students in dynamic simulations. They used nested cross-validation, a well-known machine learning approach that entails performing sub-cycles of cross-validation within larger cross-validation cycles to determine model parameters since avoiding contamination of the test set. Therefore, the inner cross-validation cycles are used to fit their model, while the outer cross-validation cycles are used to test their model's correctness.

To address the manner in which tips are delivered in intelligent tutoring platforms, Maniktala et al. [41] developed a tip delivery method termed "Assertions" for use in a data-driven smart instructor. They performed a clustering analysis that demonstrates patterns of constructive persistence among learners with limited previous knowledge whenever the tutor offers uninvited assistance in terms of Assertions. They also used clustering to deduce trends in our tutor's productivity, perseverance, and use of unsolicited hints. They argued that clustering research enables a more nuanced knowledge of student behaviour patterns related to productivity, engagement, and proactive hint utilisation. They conducted cluster analysis on standardised characteristics using hierarchical clustering with Ward's technique. They used a majority vote to determine the number of clusters through three factors: Silhouette, Calinski-Harabasz, and the Davies-Bouldin Index. Furthermore, Ahmad et al. [42] employed a similar approach for analysing the impacts of tailored analytics-based feedback on time management, which used agglomerative hierarchical clustering relying on Ward's algorithm.

To incorporate flexible navigation and material recommendation, Chau et al. [40] demonstrated a method for automated concept retrieval from digital textbooks using supervised feature-based machine learning. They discussed that the technique was developed to develop domain and learner models that serve as the foundation for intelligent

textbooks. They developed and tested a comprehensive feature set for machine learning to identify labelled concepts. Their feature-set encompasses both language and statistical characteristics. They automated concept retrieval from digital textbooks using a supervised machine learning model with a set of rich, precisely hand-crafted characteristics.

Alshmrany [46] suggested a style prediction system based on Convolutional Neural Networks (CNN) and Levy Flight Distributions (CNN-LFD). They begin by predicting automatic learning types and classifying them depending on the number of possible learning styles. Following them, when a student checks in with their user ID, the information is recorded in the database. Besides the order of the user's learning activity, features such as assessment score, login information, and login time are retrieved. Another illustration of the utilisation of convolutional neural networks is El Bahri. et al [47] who employed CNN to identify the personality of the learner. Their proposed solution has the potential to assist institutions in the identification of student personality. It can also help in the automatic adaptation of some aspects, such as differentiated instruction and personalised learning, by the student's personality characteristics. Recent comparative work has also highlighted the importance of systematically evaluating multiple AI and machine learning models for student outcome prediction [54]. This underscores the need for systematic evaluation when developing personalised learning systems.

Vazquez et al. [55] proposed that the educational usage of chatbots as a tool for promoting more personalised and integrable learning in a variety of virtual learning settings may transform them into a highly beneficial teaching tool and inspire widespread training and more flexible digital workspaces. Moreover, in an effort to optimise student engagement, motivation, and learning outcomes, the pedagogical framework of Pin-Chuan Lin et al. [56] incorporates personalised chatbots into self-regulated and active learning (SRAL). Their framework is designed to clarify the ways in which personalised AI chatbots can contribute to students' active engagement and SRAL, given their growing prevalence in education as a means of providing personalised learning support, enhancing student engagement, and supporting SRAL. Hooshyar et al. [43] developed a new Flowchart-based intelligent tutoring system that uses Bayesian networks for decision-making to assist students with problem-solving tasks and computer programming learning. They believed that to deliver individualised environments in a system, users' interactions with the system should be observed to create a profile for each user and update the Bayesian network accordingly.

An innovative approach that employs reinforcement learning algorithms to improve personalised learning was introduced by Osakwe et al [50]. Their framework is intended to optimise the sequence of Self-Regulated Learning (SRL) processes for a learning task. The objective is to find the most effective sequence of actions that results in the greatest improvement in learning outcomes, such as improved assessment scores or increased learning gains, by approaching the optimisation of the SRL strategy as a series of decision-making stages, each of which is associated with an SRL process. Their results indicated that the reinforcement learning model they devised outperformed both Long Short-Term Memory neural networks and genetic algorithms in terms of optimising SRL strategies to enhance academic performance predictions. Their methodology can be customised to create a structured strategy that offers personalised guidance on the most effective SRL strategy to employ during the execution of a learning task.

ElSayary [51] conducted a study to explore the feasibility of utilising ChatGPT as a supportive tool for educational purposes. According to them, ChatGPT is an advanced AI system that is currently trending and has the ability to transform learning and teaching by offering students personalised and flexible learning experiences. They also examined the views of teachers on utilising ChatGPT as a supporting instrument for pedagogical advancement in the digital age. Furthermore, Al Shloul et al. [52] investigated the use of ChatGPT in the field of education by conducting surveys and experiments to determine its impact on personalised feedback, interactive learning, and new teaching approaches. Their research finding demonstrated that activity-based learning improves student's level of involvement and ability to think critically. They also stated that students who engage in activity-based learning exhibit enhanced academic performance, which can be linked to their active participation and practical knowledge utilisation. According to them, the incorporation of ChatGPT provides new opportunities for interactive education and personalised support, promoting student's comprehension and investigation of intricate ideas. In addition, Einarsson et al. [57] performed a preliminary investigation utilising ChatGPT to adapt challenges in 17 various disciplines, which experts then assessed. Their findings indicate the considerable potential of ChatGPT in redefining challenges for multiple contexts. So, the introduction of ChatGPT has brought AI-generated content into the spotlight, making its efficient usage in education a popular subject of discussion [58]. Another example of using generative AI is by Sun et al. [53], who designed a personalised learning path system for higher education using a combination of transformer models, training of adversarial networks, and quantum-enhanced data processing. Their method integrates multimodal data like text, audio, and video to create more accurate, adaptive, and efficient learning experiences. Generative AI, led by tools like ChatGPT, is now widely used across sectors, including education, to expand creative possibilities [59].

3.2.2 Non-machine learning-based algorithms

This section discusses some research that employs non-machine learning methodologies. Several studies employed pattern recognition, image processing, and learning style models as non-machine learning strategies [28]. Table 5 shows some instances of non-machine learning algorithms that have been used to produce intelligent individualised learning.

To address issues with knowledge representation and reasoning, Perikos et al. [60] devised a technique for deriving first-order logic (FOL) formulae from natural language (NL) phrases. They designed the NLtoFOL platform to assist learners in overcoming such obstacles and outfitted it with a robust support and feedback mechanism. They demonstrated the mechanism of feedback. Further, they stated that the mechanism could give advice before submitting an answer; however, it is primarily used to provide

Table 5 Some examples of non-machine learning algorithms based for development of AI-based personalised learning

Type of algorithm	References
First-order logic (FOL)	Translating sentences in natural language (NL) with first-order logic (FOL) [60]
Mixed-Reality	Mixed-reality AI vision methodology [62]
Genetic Algorithm	Using Genetic algorithm [61]
Learning Pattern	Utilising learning patterns and data-driven approach [63]

support after an answer is posted. According to them, using a predefined answer classification method classifies the response in terms of reliability and consistency to assess the degree of incorrectness.

Natural language feedback patterns created automatically progress from broad to particular and may contain remarks about a student's cognitive strategies state [61]. The phrases are created automatically as feedback using a template-based language processing technique. Additionally, they gave a comprehensive evaluation of the help and feedback mechanisms' impact on student learning [61]. Lin et al. [61] suggested an online learning system based on genetic algorithms to help learners develop their comprehension of object-oriented programming ideas by personalising learning materials to every student's strengths and shortcomings. They developed customised learning materials based on learners' strengths and limitations in studying object-oriented programming principles using AI technology. They mentioned that genetic algorithms, which are based on Darwin's theory of evolution, are a type of AI technology that is commonly used to solve real-world issues. Additionally, they believe that genetic algorithms are fundamentally an optimisation technique for obtaining answers through an iterative process of evolution.

To determine whether effective learning is most practical when applied as guided study and practice, productive hands-on activity, or a mixture of the two, Yannier et al. [62] developed a smart science station software that utilises AI to assist children in learning science through hands-on experience. They argued that immediate personalised feedback is enabled by a customised AI computer vision system they built to monitor what students do in their physical environment while conducting experiments and making discoveries using tangible things. The proposed learning environment is enabled by their mixed-reality AI vision system, which enables the education system to examine and evaluate student's actions to precisely monitor and assess student predictions, studies, and understandings—and to provide a smart facilitated STEM learning environment for learners with diverse backgrounds and assistance.

Osipenko [63] used a non-negative matrix factorisation to identify frequent learning patterns and associated alignments from online learning information, ensuring that the resulting output was meaningfully non-negative. A data-driven paradigm has been proposed by them in which specific course interactions are linear functions of common stylised learning patterns customised for each student based on their affinities. Their model enables a coherent approximation of observed behaviour as a collection of construction blocks absorbed in the real data via personal affinities to each pattern. Moreover, they believed that the configuration might simply be adapted to another learning type model and to any accessible course interaction data by evaluating appropriate characteristics and adjusting the number of components to reflect the desired styled behaviour.

3.2.3 Hybrid algorithms

This section describes how to produce customised learning through the use of a combination of strategies. Table 6 shows some examples of hybrid algorithms used in the creation of intelligent personalised learning via AI.

Considering image processing and machine learning techniques, Sungkur et al. [64] provided an overview of the many available strategies that contribute to the enhancement of the learning mechanism by providing a real-time monitoring system. For

Table 6 Some examples of hybrid algorithms for the development of intelligent personalised learning with AI

Type of algorithm	References
Image Processing and Machine Learning tools	Image processing and technique of eye detection [64]
Machine Learning and Fuzzy Logic Technique	Combination of machine learning and Fuzzy Logic approach [65]
Big Data Analysis, 3D modelling, and 3D Printing	Combination of big data analysis, three-dimensional modelling, and three-dimensional printing techniques [70]
Support Vector Machine and Literature-based Technique	Combination of an approach based on the literature and a Support Vector Machine [66]
Machine Learning and Computer Vision	Personalising feedback with computer vision and machine learning techniques [67]
Fuzzy Algorithm and Neural Network Model	Fuzzy C Means algorithm and Back Propagation Neural Network algorithm [68]
GPS Technology and Data Mining Technology	Personalized data mining combined by GPS technology [69]
Natural Language Processing and embedding with the transformer-based BERT model, Google Gemini, OpenAI and K-means clustering	Predicting learning achievement by Natural Language Processing and embedding with some other techniques [71]
Knowledge Graphs, Graph Convolutional Networks and pre-trained transformer language model encoders	Personalised recommendations of knowledge concepts [72]

constructing the eye-tracking system, they mentioned that an infrared camera hidden beneath a display monitor was utilised in conjunction with image processing software to identify and distinguish the tracking eye's features. They also believed that the infrared light from an LED scaled on the infrared camera is projected directly into the eye, generating strong reflections. Additionally, they stated that infrared light is employed to prevent the user from being confused by visible light. Further, they noted the image processor and eye detector technique subsequently identify the eyes, the precise location of the student, and the right projections from the imaging systems and their positions to enhance learning experiences.

In order to propose a responsive tutoring platform for foreign languages, Troussas et al. [65] created a unique method for mistake identification utilising machine learning in combination with a fuzzy logic theory for adaptive learning in higher education. Amir et al. [66] employed a combination of a literature-based technique and a Support Vector Machine to determine learning styles automatically in a learning management system. They asserted that each learner has unique preferences when it comes to the learning process. Preferences vary according to the learner's learning style. They believed the customisation of e-learning is an assessment of online learning that features information that has been personalised for each learner's learning style. Further, they mentioned that detecting a learner's style of instruction requires an effective and precise approach. Their study merged a literature-based technique with a support vector machine to determine student's learning styles.

For offering customised assistance Echeverria et al. [67] demonstrated a platform that uses AI to aid with karate practice. To forecast the opponent's attack movement, they integrated deep learning computer vision algorithms such as OpenPose, which creates a

skeleton of the motion, with Weka movement indications. Additionally, they used Open-Pose facial detection algorithms to examine the opponent's emotional state throughout the kumite, hypothesising that this shift might also indicate the opponent's purpose about the next action to be completed. Kolekar et al. [68] believed that an e-learning system's main criteria are to deliver a tailored interface with customised material that adjusts to the student's learning patterns. They suggested accomplishing this by collecting the student's learning behaviour on the e-learning site through Web Log Mining. Following them, the learning styles are classified using the Felder-Silverman Learning Style Model. Further, they mentioned that clustering the collected learning behavioural data into Felder-Silverman Learning Style Model classifications is accomplished using the Fuzzy C Means method. They stated that because a learner's learning methods change over time, the system should adapt and deliver the required interface and material. This is accomplished by using the Gravitational Search-based Back Propagation Neural Network method to anticipate the student's learning patterns in real-time [68].

Yao [69], combined a context-aware technique that may be utilised to give appropriate learning data using GPS technology with a quick and consumer QR code interface that allows students to download important external resources as learning materials instantly. A multimedia learning component was integrated with a context-aware subsystem to recognise learners' relevant data and then select the most effective curriculum package for students in a variety of environments; this was followed by learning via a learning recommendation component, which was developed to assist personalised learning advice and suggestions. As a result, their system is comprised of a consumer QR code input control interface, wireless networking technologies, a position tracking mechanism utilising GPS technology, and tailored data mining methods.

Santos [70] argued that the field of artificial intelligence research in education might benefit from incorporating contemporary technology advancements such as big data processing, 3D modelling, and 3D printing. This combination may be used to target the area of psychomotor learning. They discussed that motor skill acquisition, like cognitive skill acquisition, may benefit from tailored training and assistance. They also believed that procedural learning had been explored since the earliest times of AIED up to this point. Lin et al. [71] proposed an innovative method for predicting learning achievement in online systems by employing natural language processing and embedding some other techniques. Their focus was on examining student behaviours through system logs. They used the transformer-based BERT model, Google Gemini, and OpenAI's embedding techniques to process data. By employing k-means clustering, they defined learning patterns and trained a machine learning model to predict learning outcomes. To offer personalised and transparent recommendations of knowledge concepts on the massive open online courses platform, Alatrash et al. [72] introduced ConceptGCN, an integrated framework that utilises knowledge graphs, Graph Convolutional Networks, and pre-trained transformer language model encoders.

3.3 Intelligent personalised learning applications

This section addresses the third research question by analysing the primary applications of AI-based personalised learning. Identifying all current applications enables the discovery of previously undiscovered gaps and constraints. Table 7 lists several recent intelligent, personalised learning applications with their categories and references. As

Table 7 Various applications of AI-based personalised learning have been classified into three broad categories, accompanied by references

Type of application	References
Personalised Content Recommendation	Personalised research paper recommender [45] Personalised course recommendations [24, 76] Recommendations that are explicable in a personalised programming practise system [73] Personalised curriculum development [49] Personalised learning system for computer science programmes [79] Personalized teacher recommendation [80] Video lecture augmentation using tagged community [81] Highly adaptive learning path recommendation [82] ChatGPT-based content summarization [83]
Personalised Tutoring and Support	Developing adaptive intelligent tutoring system [95] Adaptive web-based computer tutor [91] Providing personalised support [67] An intelligent tutoring system for enhancing problem-solving skills [43] Personalized learning in intelligent tutoring systems [44] Intelligent personalised tutoring system for personalising algebra learning [92] Intelligent tutoring system to enhance the diagnostic capabilities of medical students [94] Mentoring with computer support for personalized maths learning [86] Personalised tutoring for foreign languages(English and French) [65] Adaptive tutoring platform for conceptual physics [96]
Personalised Feedback and Assessment	Developing adaptive feedback model [97]. External feedback for self-regulated learning [98] Mathematics learning and assessment [77] Personalised learning and assessment [99] Prediction of student performance [100] Evaluating a persuasive engagement intervention [101] Assessment of reflective writing for personal essay genre [38] Using Chatbot to create flexible and personalised learning environments [55] Providing real-time feedback using multi-modal data [48]

illustrated in Table 7, based on a thorough investigation of numerous research articles on this field, we classify existing applications into three broad categories: personalised content recommendations, personalised tutoring and support, and personalised feedback and assessment systems. Articles were categorised according to their primary objective. Studies whose main focus was recommending resources or content were classified as *Content Recommendations*. Systems that primarily guided, supported, or assisted learners were placed under *Tutoring and Support*, while those centred on assessment, performance monitoring, or feedback-driven adaptation were included in *Feedback and Assessment*. As shown in Fig. 5, when articles combined two or more of these functions in an integrated way and placed comparable emphasis on more than one,

they were categorised as *Hybrid*. The presence of this category (16.8%) highlights that in practice these application areas are often interdependent and integrated, even though we analysed them separately for clarity of presentation. The following describes these three categories.

3.3.1 Personalised content recommendation

The widespread use of recommender systems in daily tasks prompted academics from a variety of fields to investigate recommender technologies' applicability in education. This study area has steadily grown to encompass a range of recommendation categories, ranging from proposing a discussion thread to advising the next challenge to solve to suggesting courses to undertake [73]. Rather than preferences or interests, the primary motivation for recommending material is the learner's knowledge level [73]. Barria-Pineda et al. [73] demonstrated how to create knowledge-based suggestions that are accompanied by visual and vocal explanations. They developed content suggestions for the Java customised programming practice system by conducting an investigation into explainable recommendations in the framework of a personalised practice system for students. Their online Java programming practice system offers personalised content recommendations based on the balance between the student's current level of knowledge and the opportunity to learn new concepts. To gauge the impact of these educational recommendations, they undertook a semester-long classroom study with 86 undergraduates enrolled in a Java programming class at a European university.

To customise courses and training sequences to users' unique needs, Xu et al. [24] provided a systematic procedure for presenting students with customised course sequence ideas. They developed a forward-search backward-induction algorithm to efficiently choose course sequences, resulting in a reduced amount of time required for a student to complete their degree. They utilised the multiarmed bandits tool to develop an algorithm that can optimally recommend a course sequence that not only shortens the time to graduation but also improves the student's overall GPA. The authors have clarified that the objective of the forward search is to identify the potential course states to reduce the issue's complexity. Their result of the forward search phase is an AND/OR graph in which each course subsequence is represented by a subgraph. The authors have applied the forward-search backward-induction algorithm to generate possible course sequence recommendation policies for students.

Following Wong et al. [49], students enrolled in higher education should pick relevant courses to fulfil their degree's graduation criteria. They also believed that the selection techniques are handbook guidelines, online platforms, and personalised assistance from academic experts, and a course recommendation system is one way to scale suggestions for large cohorts. They proposed a new deep learning approach, such as Long Short-Term Memory Recurrent Neural Networks, to handle this problem [49]. This form of recommendation system is equivalent to flexible production of adaptive content [74]. Mohseni et al. [45] introduced a web-based recommendation platform that combines word embedding and a sequence creator to provide students with a tailored list of scientific papers. They used natural language processing, which provides educational settings to show students with papers that are relevant to their prior knowledge and desires. They discussed that instructors frequently prioritise the presentation of learning materials based on the course's general learning objectives rather than tailoring the presentation

to individual students. Thus, presenting paper sequences with increasing concentrations of unique content can encourage learners' desire to learn more.

Following Chen et al. [75] computer-based learning provides learners with better access to educational materials and facilitates the development of tailored learning processes. These functions are applicable to both personalised and collaborative learning. Additionally, they stated that adaptive solutions would be implemented, and recommender systems could be used to personalise. Customisable hypermedia and recommendation platforms can be used to personalise the material [75]. A research paper (Benhamdi et al. [76]) set out to create a new method of customisation that would present students with the most suitable learning materials based on their desires, hobbies, prior knowledge, and memory capacity for storing information. They discussed that the New multi-personalised Recommender for Electronic Learning (NPR E-Learning) is a novel recommendation system based on cooperative and content-based filtering. Their method was implemented in a classroom environment to facilitate the delivery of individualised learning materials.

Universities promote the use of new approaches and teaching practices to create an engaging and participatory academic environment in which students are the centre of attention [77]. Increased student engagement and improved outcomes occur in such a tailored learning environment [77]. The instructor included content personalisation into the course's instruction to increase student's motivation in courses. In accordance with the discussion of Revelo et al. [78], personalisation of content is intended to increase student engagement and learners' confidence in applying newly acquired abilities in their lives and jobs. Further, they stated that students participated in the lesson by suggesting applications for the subjects discussed. Their concepts are then incorporated into lectures, home assignments, and assessment difficulties. Additionally, they argued that during the term, the curriculum is changed to accommodate student's connections to digital information in relation to their educational, personal, or industrial interests. Customisation of content is a more personalised variant of course tailoring [78].

Personalisation and flexibility have become critical characteristics of educational technology as a result of the fast advancements in e-learning systems [79]. Aeiad et al. [79] examined the process of developing an architecture for a personalised and adaptable e-Learning system, which aims to progress the discipline. They believed that a flexible e-Learning system's goal is to give users a customised and adaptable learning environment based on publicly available Web resources. Their learning subject was modelled using a taxonomy, and appropriate learning resources were extracted from the Web using a learner's model. Their system uses natural language processing techniques to compare material collected from available resources to a range of learning objectives established by the standard curriculum to facilitate the subject's proper learning. In their system, a computer science application is utilised to demonstrate the personalised e-Learning system's operation on their platform. Chen et al. [80] developed practical teacher recommender systems specifically for K-12 one-on-one online classrooms. The authors also highlighted challenges associated with teacher recommendation systems, such as limited supply and demand, resulting in a restricted number of students that teachers can accommodate per semester. They proposed a personalised recommendation system for teachers that could create strong pseudo-training labels, which would serve as a reliable indicator of optimal teacher-student matches. Additionally, they

suggested providing incentives to new teachers to enhance their ranking scores in the recommended candidate pool.

Recent work by Ghosh [81] introduced an AI-powered system that improves video lectures by integrating tailored supplementary resources derived from community question-and-answer platforms. By automatically determining key course concepts within lecture content, the system retrieves and categorises relevant question-and-answer pairs to help with a deeper understanding. Students can engage with resources that are specifically tailored to their needs by tagging these augmentations according to their pedagogical roles, such as conceptual clarification or tool usage. Particularly for motivated students who engage in self-directed study, the hierarchical interface and content personalisation were demonstrated to enhance user satisfaction and learning outcomes substantially. Moreover, Luo et al. [82] suggested a personalised learning path recommendation model that dynamically tailors content delivery based on each student's current knowledge state and learning objectives. The system considers the logical learning subject sequence and recommends a sequence that minimises cognitive load while encouraging even knowledge growth. Unlike traditional techniques, it introduces a decision-making framework that optimises learning paths by balancing three competing objectives-learning performance, time efficiency, and difficulty progression-ensuring that the recommended content remains goal-aligned, time-conscious, and cognitively manageable throughout the learning process.

Nguyen et al. [83] investigated how AI-powered text summarisation tools like Copilot, QuillBot, and ChatGPT assist students in retrieving relevant content more efficiently. By providing summaries to different text types-such as scientific, technical, or literary-the study emphasises how these tools can adapt information presentation to suit individual learners' academic requirements and preferences. This adaptability improves students' ability to engage with complex material in a time-effective and personalised way, supporting more autonomous and targeted learning experiences. Ng et al. [84] further examined the integration of generative AI tools in the education system, highlighting both the opportunities and institutional challenges in supporting personalised learning strategies through teacher training, infrastructure, and policy frameworks. Similarly, Samala et al. [85] developed an extensive taxonomy of generative AI applications in education, emphasising the increasing significance of models such as ChatGPT in personalised learning and content generation while also emphasising the necessity for ethical considerations and pedagogical protections.

3.3.2 Personalised tutoring and support

A major randomised control experiment found that tailored human tutoring may considerably improve arithmetic proficiency for minority students living in high-poverty communities [86]. Unfortunately, these increases came at a significant resource cost; with just two students each class time receiving instruction from a tutor, the additional expenses per student are not practical in many districts [86]. The top three research topics in open learner models are intelligent tutoring systems, self-regulated learning, and learning analytics [87]. One critical function of this type of learner model is to automatically personalise instruction or recommendations for the learner [87]. When compared to traditional educational experiences, intelligent tutoring systems have been found to improve student learning results [88]. Similarly, adaptive learning and recommendation

algorithms are becoming offered in a growing number of online learning settings [88]. When designed effectively, these sorts of systems obviously benefit students by improving the approach to learning a well-defined domain with a formally modelable curriculum and ways of reasoning [88]. Additionally, students reflected that customised e-learning platforms could help them personalise their learning process depending on their techniques and keeping to the class pace [89]. The tutor's hint's context sensitivity is an example of adaptable and flexible behaviour [90].

Following Schaldenbrand et al. [86], while personalised learning that is targeted to each student's social, cognitive, and organisational requirements is the ideal answer, it is not financially or logistically feasible for every student to acquire one-by-one supervision from a human mentor. Thus, although computerised tutoring demonstrates that more possibilities for learning result in increased accomplishment, they also highlight that there are disparities in the availability and types of choices available to underprivileged students that innovation alone cannot address. They also believed that personalised learning with tutors has the potential to address this opportunity gap but is also expensive to deploy. Schaldenbrand et al. [86] offered a personalised software intended to increase tutoring efficiency by proposing learning materials selected from the internet to tutors based on their learners' use of educational technology applications. Their software's ability to categorise students based on mathematical learning program data aided mentors in concentrating their attempts and coaches discovered resources to significantly enhance their teaching and mentoring practices. Additionally, their proposed platform includes capabilities for quickly accessing various resources and connecting them to the requirements of students and tutors.

Aleven et al. [91] worked on invention activities while developing adaptive support, which are inquiry-based activities in which learners construct representations that explain facts provided in contrasting circumstances. They developed and piloted a specialised adaptive computer coach to assist instructors in implementing these activities in their classrooms. The Mentor has been developed by them as a model-tracing coach, with rule-based modelling encapsulating its pedagogical coaching technique, which was developed in part using data from human tutors. They discussed the invention coach and the instructional paradigm that accompanies it.

According to Kochmar et al. [44], compared to other computer-based educational techniques, intelligent tutoring methods have been much more successful in promoting learning. On the other hand, they argued that numerous intelligent teaching systems mainly rely on human design and custom-crafted rules, complicating their development and transmission across domains and limiting their potential usefulness. They also believed that the development of a machine learning technology could assist in the automatic development of individualised feedback that considers student's unique needs while removing the need for expert interaction and the formulation of hand-crafted criteria. They argued that students could receive tailored feedback using machine learning and natural language processing approaches. Additionally, their individualised feedback system results in increased success rates while solving practice activities. Their individualised feedback platform results in a large overall improvement of 22.95% in student achievement outcomes and significant progress in subjective feedback appraisal.

Walkington et al. [92] demonstrated how an Intelligent Tutoring System for mathematics might be used to customise learning by linking to individuals' specific interests.

They investigated the hypothesis that the impacts of personalisation may be mitigated by student's quantitative participation in extracurricular activities. They tested whether arithmetic problems that drew on student's deep or superficial knowledge of their unique interests had differential impacts. Their findings implied that tying math education to student's extracurricular activities might benefit learning in an Intelligent Tutoring System and prevent system gaming. However, advantages may be obtained only when student's quantitative involvement with their extracurricular activities is comparable to the depth of the individualised difficulties. Moreover, they stated that students who have a low level of quantitative engagement with their hobbies might gain the most from tasks that focus on superficial parts of their interests. They believed students who demonstrate a high level of statistical engagement with their hobbies might gain the most from challenges that thoroughly include their interests in the mathematical structure. They discovered that issues involving greater personalisation could promote favourable emotional states and ward off negative ones in all learners.

Huang et al. [93] examined the concurrent presence of self-regulated learning and medical reasoning in medical students as they worked through a diagnostic task in an intelligent tutoring system. They argued that, at last, students could receive personalised feedback on various factors, such as the relevance of lab tests to the correct diagnosis, accuracy, and the coherence of evidence with specialists. They aimed to offer novel perspectives on the performance disparities between students with high and low achievement in tasks of variable complexity. In a similar way, Wang et al. [94] designed an intelligent tutoring system to enhance the diagnostic capabilities of medical students. They sought to investigate the impact of computer-based scaffolding on the efficacy of problem-solving and metacognitive monitoring among students. In order to clarify the advantageous effects of metacognitive scaffolding on metacognitive monitoring, they employed the framework of cue utilisation.

Artificial intelligence approaches in computer-assisted learning languages create a more personalised and adaptable educational environment [65]. Troussas et al. [65] demonstrated an intelligent teaching method for the acquisition of English and French ideas. Their system contains a revolutionary machine learning-based approach for error diagnostics. Their approach used two computational techniques, namely Approximate String Matching and String Meaning Similarity, to identify spelling errors, tense errors, supplementary verb errors, and errors resulting from language teaching problems. They used the fuzzy logic approach for error diagnostic testing, which inputs the outcomes of the first step or the knowledge interdependences between the various domain ideas in the training materials and adaptively decides on the most appropriate learning material supply to the student each time. In addition, utilising meta-synthesis techniques to study the findings of prior research can aid in the examination of adaptability aspects and the intelligent tutoring system [95].

Echeverria et al. [67] established an AI-based personalised system called KUMITRON to assist with karate practice to provide adaptive support. They argued that KUMITRON, in particular, continuously monitors the karatekas' physical activity throughout Kumite (Karate combat) to use a variety of sensors (physiological and inertial) and a drone camera to supply multisensory personalised feedback approximated using machine learning and computer vision techniques. Their integrated monitoring seeks to predict a fighter's

mat motions, adding value to training and delivering individualised recommendations on the best approach to employ in an attempt to win the Kumite.

3.3.3 *Personalised feedback and assessment*

Since feedback has a significant impact on learners' ability to enhance their learning outcomes, this could be as closely linked with the attributes of each student as feasible [97]. The accuracy of a learner's response is commonly used to predict the learner's characteristic, with the correct response regarded as expressing the learner's understanding, while the erroneous response is interpreted as showing the learner's confusion. In particular, wrong responses show that learners seek assistance in resolving their misconceptions [97]. Great teachers can provide personalised feedback and appropriate challenges without technology; however, technologies can make this kind of personalisation easier to manage and deliver [102, 103]. External feedback has also been recommended as a way to improve student's self-regulated learning [98].

Due to student's varying degrees of knowledge, they should receive differentiated feedback in the same manner that varied levels of accuracy receive differentiated feedback [97]. Personalised feedback on confidence information tries to bolster learners' confidence who possess an accurate view but lack the self-confidence to remember it [97]. Moreover, adaptive feedback tries to decrease learners' confidence in their misunderstandings and subsequently correct them in their knowledge [97]. Pailai et al. [97] suggested a new feedback system for the Kit-Build conceptual framework with confidence tagging that would deliver personalised feedback based on accuracy and reliability data. They argued that the Kit-Build concept map is a digital product that complements the approach for idea maps. For mathematics education and examination, Azevedo et al. [77] introduced the MathE platform, which aims to foster and accelerate mathematics learning in higher education. To ascertain student's and teachers' perceptions of the collaborative electronic learning mode, their MathE platform also solicited user feedback. They stated that subject-matter training is integrated with reading and writing abilities in a variety of ways. Additionally, they said that learners acquire their ability to read, understand, and write documents in discipline-specific genres. However, writing may be utilised for more than just practice in communication skills; it could be used to analyse previously learned content [77].

Klebanov et al. [38] addressed a writing intervention and the utility value intervention. That has been found to be beneficial in laboratory and field trials for increasing interest and retention in STEM topics. They provided an in-depth examination of the possibilities for natural language processing tools to facilitate large-scale appraisal of such writing. They also developed a set of characteristics that distinguish utility value writing across genres, highlight common themes, and test utility value scoring models using essays on well-known and novel biological topics. Further, they argued that the automatic utility value score findings are encouraging, particularly for the category of personal essays. Si [104] believes that learners develop language and communication skills in the setting of social interactions inside their language training environment. Their proposed method established and guided these encounters through the use of software virtual settings and virtual personalities.

According to Kong et al. [105], the learners were assigned to formative evaluation activities such as reading discussions, exams, demonstrations, and cumulative evaluation

strategies, including writing essays for a school-based personalised learning program. They argued that the teacher can give real-time face-to-face feedback on group work material and online group conversation topics. They believed that using a networking platform with an online course enables the teacher to supervise and monitor student's academic progress, as homework, quiz scores, discussion, and other interactions could all be documented [105]. Regarding online courses, massive open online courses provide learners with plenty of opportunities to access courses that cover a diverse array of disciplines within higher education at no cost or at a reasonable cost [106]. Moreover, Kim et al. [107] discussed how the student mental model analyser for research and teaching. They used a web-based formative feedback and assessment technology as an instructional system to assist students in writing strong summaries that aid in their comprehension of complex texts and manage them for meaningful classroom discussion. As an illustration, one student can suggest creating a music synthesiser, described by Revel et al. [78]. They discussed that the teacher offered adaptive feedback to the learner, stating that they employed waveform synthesis or developed devices based on a finite-state system. As a consequence, the student created a keyboard synthesiser using circuitry. Further, they said that the learner did not create the synthesiser but followed directions from an Internet-based example project. The end-of-semester demonstration was great, and the student displayed a thorough understanding of the oscillator ideas and formulae presented in the course [78].

As Kuo et al. [108] described, the conventional lecture-based method of instruction can no longer encourage student's creativity and integration into society. They believed that without motivation and excitement for learning, students might have had a strong propensity to decline to engage in classroom activities. To excite and motivate students in learning, an increasing number of educators have shifted their teaching methods from instructor to student-centeredness. In addition, they argued that learners create analytical thinking, problem-solving, and effective communication through an active learning strategy that incorporates discussion, interviewing, and prototyping. Through interaction with others, students can develop their drive and social skills [108].

A well-defined idea of customised learning can act as an umbrella term for teaching approaches that aim to take into account each student's unique talents and learning requirements [109]. Schmid et al. [109] examined the theoretical features of this idea and the in-school usage of digital technologies for individualised learning as reported by learners concerning their self-assessed digital abilities and attitudes about the utility of information technology in education. Moreover, they said that the improvement in reported use of digital technologies in education systems characterised by open teaching techniques is an essential part of personalised learning. According to a conceptual model, it has a beneficial impact on self-reported technological knowledge and self-perceived perceptions in learning [109].

There is growing interest in learning technology in gathering and integrating data from a variety of sources and devices in examining learning progress [48]. Mitri et al. [48] described a cardiopulmonary resuscitation (CPR) tutor, a multimodal feedback platform for real-time CPR training. They believed that The CPR tutor utilises recurrent neural networks to recognise training errors. Additionally, the CPR tutor automatically identifies and grades the accuracy of chest compressions using five CPR performance markers. Their proposed platform analyses a mixed data stream comprised of kinematic and

electromyographic data in real-time to discover training errors. In addition, they argued that the CPR teacher delivers audio feedback based on this evaluation to help address the most crucial errors and improve CPR performance. The CPR tutor's error detection algorithms were developed by them using data from ten experts. Their proposed model can provide real-time feedback based on multimodal data and can analyse field research findings by quantifying the influence of CPR tutor comments on evaluation criteria and qualitatively assessing student questionnaire responses.

While offering personalised instruction in a large classroom is an efficient technique to boost teaching quality, the traditional teaching style makes it challenging for teachers to become acquainted with each learner's learning circumstance [100]. Blended learning enables teachers to provide personalised instruction. Xu et al. [100] have studied research on integrating a flipped classroom with a small private online program to execute blended learning and explore personalised education. Their study also examined the predictability of student's academic performance in this strategy. They employed the K-nearest neighbours method, a form of machine learning, to validate the generalisation of the prediction. They argued that students' behaviour could be utilised to predict their learning results in blended learning, and the deployment approach is replicable. They also believed that in blended learning, teachers could offer personalised training. Suppose simply online learning behaviour can be used to predict student success. In that case, instructors may save time and energy gathering offline information and make full use of learning analytics to support individualised teaching in blended learning [100]. So, teachers' utilisation of learning analytics to shape adaptive and individualised learning is fast-growing [110]. Learning analytics has a unique position that allows it to utilise and perhaps make contributions to philosophical and epistemological topics [111].

Following Wanner et al. [99], in higher education, the words "flipped classroom" and "adaptable teaching and learning" are prevalent. They discussed that students and learning are at the centre of pedagogy and practice in higher education, which has altered significantly over the last several decades due to new technology and the growing availability and use of face-to-face, online, and blended courses. To provide a consistent educational experience in higher education, assessment techniques must receive the same amount of attention as personalisation and learning flexibility [99]. Moreover, Wanner et al. [99] presented an investigation of the utilisation of a flipped classroom and customisable assessment components. Following them, students feel and are more motivated in blended learning, prefer hybrid learning to entirely online learning, need and demand a proper framework and guidelines, and place a high premium on flexible evaluation via increased choice and control.

According to El-Sabagh et al. [112], adaptive e-learning offers significant pedagogical benefits because it fosters student learning, increases students' motivation, and facilitates knowledge acquisition. We created Fig. 7, based on the findings from [112–114] to illustrate some of the engagement characteristics and advantages associated with online adaptive learning. Adaptive domain modelling entails advanced techniques for assigning appropriate learning activities and units to students to facilitate more subjective learning [112]. The more flexible the learning material is to the learners' cognitive needs and capacities, the more successful and precise the content's teaching potential becomes [112]. For instance, when studying a domain idea, if a learner with a high level of knowledge obtains a high amount of learning experiences that are inappropriate for

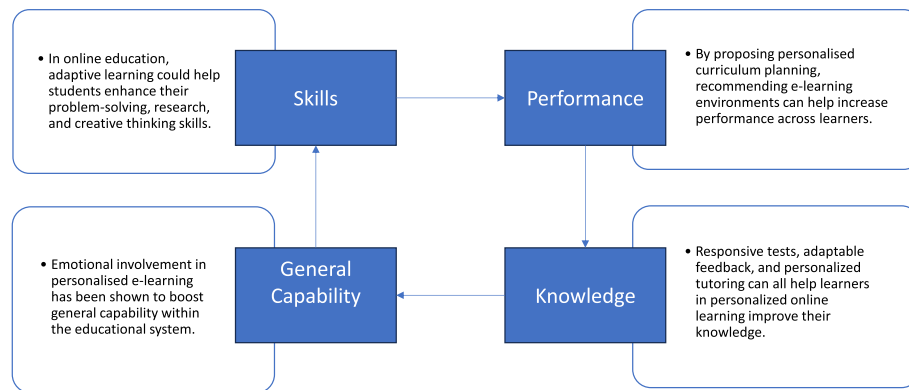


Fig. 7 Engagements factors in online personalised learning applications

Table 8 Some samples of percentages of overall improvement of student's learning, including increases in student performance, knowledge, engagement, and speed using personalised learning with AI

References	Improvement	Type
Customised Laboratory Learning for Undergraduate Electrical Engineering [119]	29%	Student Satisfaction
Using Dynamic Student Modelling for Intelligent Tutoring Systems [96]	27%	Student's Speed
Intelligent Tutoring Systems using Personalised Pedagogical Interventions [44]	22.95%	Student Performance
Using Personalizing Learning Material in an E-learning System [120]	8–20%	Student's Score
Personalised Laboratory Experiential Learning [119]	15%	Student Performance
Personalized Mobile Learning for Principle of Digital Literacy [121]	9.2%	Student Knowledge

his or her level, the learning process may provide unsatisfactory outcomes [115]. Appropriate user modelling techniques are also important to assist these approaches in meeting the demands of learners in real educational facilities [116]. Moreover, Gardner et al. [117] present an analysis of the development of AI for formative evaluation educational assessment by focusing on two key applications: automated essay scoring platforms and computerised adaptive exams, including the big data analytic techniques to machine learning that drive it.

3.4 The influence of this technology on the education system

This section discusses the findings of experiments conducted in some state-of-the-art publications on AI-based individualised learning. The findings confirm that utilising AI to personalise learning could help students enhance their knowledge and engagement. Table 8 illustrates some of these improvements, including increases in student performance, knowledge, engagement, and speed. Additionally, the most often utilised subjects for individualised learning have been considered in this section.

3.4.1 Enhancement of student's learning

According to our review, some research indicates that deploying AI-based personalised learning improves students' knowledge, abilities, satisfaction and saves their time. Kochmar et al. [44] used machine learning and natural language processing methods to

deliver individualised feedback to students through hints and Wikipedia-based descriptions. They reported the findings of studies with students and demonstrated that automated, data-driven, individualised feedback resulted in a large overall increase of 22.95% in student performance achievements and major improvements in personal feedback rating. Kochmar et al. [118] argued that their proposed platform (Korbit) trials demonstrate that extracted and created Wikipedia-based explanations result in equivalent student learning improvements. 83.33% of the time, students regarded one or both types of explanations as useful. In a real-world production setting, Chen et al. [80] applied their approach by monitoring the average number of times a student necessitates a change of teacher on a continuous basis. They discovered that the average number of matching attempts reduced from 7.22 to 3.09 throughout a five-month observation time (2020/01–2020/05), indicating that their system can offer more reliable suggestions to teachers. As another actual illustration, Barria-Pineda et al. [73] conducted a classroom study over a semester with the aim of assessing the impact of explainable educational recommendations. The study involved 86 undergraduate students who participated in a Java Programming course at a European university. The primary goal of the study is to improve the student's understanding of the target concepts and emphasize the importance of having sufficient knowledge of the prerequisite concepts required to solve the activities accurately.

Katz et al. [96] argued that several intelligent tutoring systems, which handle problem-solving domains, rely heavily on student modelling. They found that learners in the dynamic student modelling group completed conversations approximately 27% faster than students in the static learner modelling group. Pailai et al. [97] described responsive feedback of a Kit-Build concept map with confidence labels for the purpose of increasing learners' comprehension in a reading situation, and their results indicate the enhancement of learners' academic achievement and retention of knowledge. Following Al-Ataby et al. [119], students frequently express dissatisfaction with the quality of delivery in terms of the assistance they receive from classroom teachers and technicians and the absence of relevant and personalised feedback. To address this issue, Al-Ataby et al. [119] executed a framework of transformational change over a five-year period via the use of a variety of creative ways to improve the learner experience. Furthermore, they argued that integrating tools such as video tutorials, online pre- and post-lab questionnaires, and regular FAQs was particularly creative and effective in its implementation, resulting in a win scenario in which both students and teachers could communicate instantly successively in ways previously unimaginable. They mentioned this is especially important given the continued growth in student enrolment, which indicates that such tactics will be increasingly utilised. Their results show that student satisfaction has increased steadily and quantifiably over the three academic years (2012–2015) by 29%. Moreover, they reported that this resulted in an overall improvement of roughly 15% in learner performance across three academic years.

Christudas et al. [120] presented a customised learning approach to enhance the quality of self-learning in a responsive e-learning system by giving the most appropriate content to each student. They achieved this by modifying a genetic algorithm known as the Compatible Genetic Algorithm. Their findings indicate that learners' scores and satisfaction levels improved as a result of the evaluation. They stated that most student's scores increased, with 50% of learners experiencing a rise of between 8% and 20% in

their results. Yao [69] stated that their suggested method was validated experimentally and had the potential to boost English learning. Their instructional strategy was useful, particularly for students who had difficulty acquiring and using English in daily life. It has been mentioned that the independent t-test findings reveal that the experimental group's average post-test grade was greater than the control group's. Following them confidence in verbal abilities and vocabulary understanding was also greater in the experimental group than in the control group while reading English writings. Additionally, questionnaire responses suggested that the experimental group's students demonstrated more accurate and practical memorisation, especially for English vocabulary, expressions, and idiomatic than the control group's students. Furthermore, they stated that almost 80% of the users expressed satisfaction with the system's educational benefits and simplicity of use.

Hongthong et al. [121] utilised a pre-test and post-test score to determine the capacity to boost learning. They presented two sets of examinations to students, one on digital literacy concepts and another on information and cyber security. Their suggested m-learning may greatly improve these criteria by 9.2% and 6.4%, respectively. Following them, however, students can have a greater understanding of the notion of digital literacy than they can of information and cyber security. They also mentioned that the explanation for this might be that the topic is too difficult for the students to comprehend. Additionally, they requested students to complete a survey form to ascertain their level of satisfaction. Their findings indicate that the majority of students (95%) believe that this established m-learning may assist them in acquiring knowledge swiftly and conveniently. Additionally, Zhang et al. [122] provided valuable insights on the influence of AI on teacher education, emphasising the potential advantages and difficulties of incorporating AI into educational structures. Their study revealed statistical links between using AI technologies and diverse educational results. AI significantly improves classroom management, fosters an inclusive learning environment, and advances digital literacy. Furthermore, it benefits the recognition of student's learning preferences and enhances the rapport between teachers and students. The results of this study are supported by statistically significant path coefficients, demonstrating the strong impact of AI on certain educational factors.

3.4.2 Increasing positive emotional and cognitive engagement of students

Later years have seen an increase in the number of publications examining student affect and emotion in various circumstances, where students often encountered a range of emotions that directly affected their learning performance [123]. Emotions are a result of a person's experiences and responses to current circumstances; they are one of the most contentious issues topics in psychology and play a critical part in the human educational process [124]. The subject of cognitive experience encompassed statements that alluded to aspects of acquiring new knowledge or abilities [125]. Particularly in the context of online learning, students with learning disabilities encounter cognitive challenges that are likely to impact their academic performance [126]. The AI framework can enable educators to personalize educational methods based on student's unique cognitive profiles, and AI-enhanced strategies can reduce the negative effects of learning disabilities on academic outcomes [127]. Moreover, several personalised learning systems make

suggestions based on considering human emotions, attitudes, and other physiological factors [128].

To assist automated systems in determining student's emotional responses, Aslan et al. [129] gathered self-labels from learners as they analysed their internal states may be a method for collecting subject-specific data required for developing tailored emotional engagement models. Alyuz et al. [130] believed that intelligent tutoring systems are restricted in their ability to provide fully tailored learning experiences without precise identification of emotional states. As a result, they built an empathetic autonomous 'tutor' capable of intensively watching students in real-time and analysing their affective states associated with emotional involvement. They gathered 210 h of data from 17 genuine classroom trials using a webcam and context performance. They suggested a method for accurate emotional engagement detectors based on the hierarchical semi-supervised model modification. Their experimental findings indicate that personalisation improves the performance of general emotional engagement analysers. Their suggested approach of semi-supervised hierarchical customisation results in 89.23% and 75.20% F1 values, respectively, for the teaching and assessment parts.

Tsybulsky [125] presented student's cognitive and emotional responses to the training program of digital curation, as well as examples of how curation enhances individualised learning. Their study revealed that when learners expressed their cognitive and individualised educational experiences, they pointed to the overall curation process, and when they expressed their emotional learning experiences, they referred to specific curation tasks. Mayer [131] proposed an effective multimedia system in e-learning for the cognitive theory of multimedia learning to assist individuals in learning better via computer-based teaching, incorporating words and visuals rather than just words. They believed that the intricacy of the content for the learner might trigger cognitive processes aimed at mentally representing the provided material. To increase student satisfaction levels with e-learning systems, Christudas et al. [120] presented a tailored learning model based on learner behaviour that forces learning materials to be compatible. Their approach can distribute material in e-learning systems based on the students' level of satisfaction. Their findings reveal that the criteria identified for customisation result in an increase in the learners' scores and satisfaction levels. In addition, Orji et al. [101] mentioned that their study's findings also demonstrate that personalizing persuasive methods to the receptiveness of certain students multiplies their influence on engagement.

In summary, the evidence reviewed in this section shows that AI-based personalised learning has a greatly positive effect on education. Studies report gains in student performance, knowledge, and engagement, as well as improvements in satisfaction and efficiency of learning. These outcomes have been observed across different educational settings, ranging from K–12 classrooms to higher education and online learning environments. Although the methodologies and technologies may differ, the results collectively indicate that AI-based personalised learning can improve learning outcomes and offer practical ways of supporting more individualised education.

3.5 Discussion

In general, personalised-learning approaches strive to modify the learning speed and the educational tactics, material, and activities employed to the individual learner's

Table 9 Advantages of personalised education systems

No.	Benefits of a personalised learning system
1.	Empowering students with a sense of ownership over their learning and delivering real-time feedback [4]
2.	AI-based personalised learning can contribute to the diversification of the educational environment and the development of relevant fairness models [14]
3.	Students in adaptive learning have the option of selecting issues to work on and submitting their solutions for evaluation at any time and from any location [5]
4.	Customized learning can help learners from different ethnic and cultural backgrounds, with or without disabilities, receive an appropriate education [141]
5.	Personalised learning could be significant for students' engagement with online learning during the COVID-19 Pandemic [142]
6.	By enabling learners to engage, personalised learning can aid in the absorption and diffusion of knowledge and stimulate the establishment of autonomous networks [14]

strengths, limitations, and interests [4]. Moreover, the fast growth of Generative Artificial Intelligence (GenAI) models, like ChatGPT, has drawn significant attention in educational research due to their potential impact on teaching and learning practices [85]. This section has summarised the benefits and opportunities of personalised learning for both learners and academic staff. In addition, along with several possibilities, the exciting model's challenges have been discussed.

3.5.1 Opportunities for personalised learning

The advancement of customised learning research demonstrates that as technology improves, personalised learning becomes more integrated with technology and makes use of the opportunities that technology can provide [4]. As noted in the above sections, a personalised learning platform can provide a number of benefits for the educational system. Apart from the findings in the result section, which confirm that employing AI to personalise learning would help students improve their knowledge and engagement, Table 9 highlights several other benefits.

Following Maghsudi et al. [14], education has a variety of distinct effects on the well-being of individuals and society as a whole, both in the long and short term. As such, fairness is a critical component of education, which could be achieved through modern personalised learning. They also believed that diversity is also a paramount concern and it is clearly recognised in today's workplace that a diverse staff encourages new ideas and enhances efficiency. Education's social responsibility requires more than merely drawing a diverse variety of persons from diverse origins. For instance, an AI-powered personalised education platform may help diversify the educational environment by encouraging collaborative learning across several networks [14]. As noted by Xie et al. [132], personalised learning has long been an engaging topic in the field, with key data sources, such as students' preferences, academic performance, learner profiles, and activity logs, becoming essential components to support personalised experiences.

3.5.2 Challenges for personalised learning

In recent years, various studies have been undertaken on AI-based tailored learning. Apart from the numerous opportunities, there are still a number of limits and difficulties associated with these models. By introducing these challenges, researchers may overcome present limitations and generate substantial discoveries. This section discusses

two significant challenges: managing time constraints in customised learning systems and dealing with large amounts of data in these systems.

- *Time restrictions from formal learning to informal learning:* Regarding Chen et al. [133] traditionally, the major method of acquiring knowledge was via formal training in schools and colleges. They also mentioned that students spent around 90% of their daily study time on formal education and relatively little on informal education. However, as digital technology has advanced, student's learning patterns and preferences have changed. They believed nowadays, students are not entirely reliant on schools or colleges to acquire information about a certain subject. Learners can access and use informal learning techniques over the Internet. According to them, as a result, the user's available time will be restricted in personalised learning systems, which are a kind of informal learning.

According to Nabizadeh et al. [134], one of the primary challenges with personalised learning systems is for learners to achieve a route that they can successfully complete in a timely manner. Following them, present path personalisation systems need users to spend a set amount of time learning a path, which is generally fixed and determined by the method. They also mentioned that complying with these criteria and offering such pathways becomes more difficult when a user lacks the time necessary to learn a path. They believed due to a variety of factors, a user may be unable to devote sufficient time to following a whole course and several of the primary reasons include multitasking, laziness, and poor time management. Therefore, a customisation approach should consider the consumers' time limits [24, 135]. A Course planner [135], which was introduced in 2016, is capable of generating pathways based on deadlines. Given a user-specified deadline, this approach constructs a route consisting of a succession of semesters [135]. Each semester offers a range of programs that the user must take depending on each course's length [135].

- *Learning data:* In its various forms, education has long relied on data to plan, manage, and educate, and smart technology now provides new chances to broaden the data focus [136]. However, it is critical to highlight that the data obtained for various modes of delivery may differ according to technology, environment, institutional features, and instructional methodologies [137]. As a consequence, integrating data across several scenarios, developing data-centric smart learning, and providing learners with a smooth learning experience and personalised services is also a significant problem [138]. According to Nabizadeh et al. [134] as more students and learning objects are added, the quantity of data utilised as input to learning customisation approaches grows. Further, they mentioned as a result, the extent of the saved learner interaction data might become rather huge. Regardless of the volume of data, path personalisation strategies seek to respond rapidly in order to maintain user engagement. Hence, a significant challenge is developing scalable algorithms capable of dealing with big datasets [134]. Techniques for analysing large amounts of data that take into account their volume, diversity, velocity, uncertainty, and truthfulness [70].

Collecting and utilising this learning data while following privacy and data protection principles is particularly challenging. Learning analysis may be used to perform data about student progress, track progress, and provide responses to the system, educators, and students [139]. However, present research on the formulation and

construction of learning analytics is mostly driven by academics in the fields of technology and decision sciences [139]. The emphasis is on the use of insights to teaching and learning from a technology standpoint rather than a pedagogical standpoint [138].

A balanced approach to AI integration in education requires institutions and policy frameworks to go beyond technical progress and address ethical and practical dimensions. This involves equipping educators with training to use AI effectively while managing risks, enforcing strong privacy protections, and reducing algorithmic bias [140]. Moreover, creating collaborative spaces where teachers and students jointly shape how AI is used can help maintain its role as a supportive tool rather than replacing human-led learning [140].

- As a result of examining various studies on AI-based personalised learning, we can highlight the importance of three primary components in data collection:
 - *Right type of data*: Collecting data from a variety of sources such as text, audio, photos, and video can aid in gaining a better knowledge of students and providing them with a better service.
 - *Right volume of data*: Having an adequate quantity of data might be crucial in offering a more reliable, tailored learning system.
 - *High-quality data*: Having noise-free and trustworthy data is also critical for constructing a trustworthy customised learning system.

3.5.3 Limitations

While this review provides a comprehensive synthesis of recent advances, a few limitations should be noted. The scope was limited to studies published between January 2015 and June 2025 and indexed in selected databases, which may mean that some relevant work was not included. In addition, the diversity of research methods and educational contexts made it difficult to apply uniform criteria for comparison. These factors are common in systematic reviews and do not diminish the overall insights, but they do highlight the value of future studies that examine AI-based personalised learning across a wider range of settings.

3.5.4 Future research directions

Beyond descriptive insights, the synthesis of 125 studies also points to several areas where further exploration could be valuable. First, while many studies employed supervised and unsupervised machine learning for content recommendation and feedback, relatively fewer investigated reinforcement learning or generative models, even though these approaches could be relevant for managing adaptive and time-constrained learning paths. Second, studies in higher education were relatively more common than those in K–12 settings, indicating that additional research in school contexts could provide useful complementary perspectives. Third, relatively few studies focused on particular student groups, such as learners with disabilities, even though considerations of fairness and inclusion are often noted in discussions of personalised education. Future work that addresses these areas could broaden the scope of evidence and support more equitable and pedagogically grounded applications of AI-powered personalised learning.

4 Conclusion

This study systematically reviewed and analyzed a large body of literature on AI-based personalized learning using specific inclusion and exclusion criteria. The findings support the assertion that personalized learning has evolved into a core concept in educational technology research. Regarding our analysis, while customised learning is now feasible, it remains one of the most significant matters in contemporary educational institutions. This research work was undertaken by conducting a review of the literature on progress in applying AI to personalised learning, covering publications from January 2015 to June 2025 from prestigious journals and conferences. The primary objective of this research was to investigate four significant outcomes. The first outcome focused on exploring the function of AI-based personalised learning within the education system, highlighting gaps and limitations, including the challenge of insufficient data and suggesting solutions, including the usage of cutting-edge AI technology, such as AI generative models. Second, after comprehensively examining various research articles on AI-based personalised learning applications, we have classified the existing applications into three broad groups: personalised content recommendations, personalised tutoring and support, and personalised feedback and assessment systems. By classifying and analysing current personalising learning applications for learners and academic staff in various scenarios, previously unrecognised gaps and limitations could be discovered. Table 7 shows the summary of this classification. Thirdly, the novel utilised algorithms, approaches, and tools for developing an AI-based personalised learning model have been addressed in three categories: those that employ machine learning, those that do not, and the hybrid category. Finally, based on the findings from the publications, in the result part, we indicated that individualised learning has the potential to enhance educational systems on both the learning and emotional engagement fronts. Table 8 illustrates some of these improvements based on the percentage and type of enhancements.

Finally, we discussed the potential benefits and drawbacks of integrating personalised learning. The components for personalized learning offer rich research opportunities, and our results highlight the exciting potential for new discoveries. In the future, a unifying technique and framework may be built by examining the components of many customised learning systems. As we better understand cognitive emotion and build new technologies, future studies exploring the components of various individualised learning techniques will need to see this as a fluid term.

Author contributions

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